UNIVERSITY OF JOHANNESBURG

Lecturer's remarks
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Exercise 1: SQL Fundementals

Database Employees_db

Id	First name	Last name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018/06/15	NY
2	Jane	Smith	HR	48000	2019/07/20	Chicago
3	Mike	Johnson	Finance	60000	2017/01/30	LA
4	Sarah	Brown	IT	53000	2021/03/25	NY
5	David	White	Marketing	52000	2016/04/10	San Francisco
6	Emily	Davis	IT	62000	2015/02/14	Chicago
7	Robert	Wilson	Finance	59000	2012/10/01	Huston
8	Jessica	Moore	HR	51000	2018/05/22	LA
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago
10	Laura	Hall	IT	50000	2020/08/10	San francisco

Questions

Select Statement

Write a SQL query to retrieve all Columns from the employees table

```
SELECT *
FROM employees;
```

Select Distinct Statement

Write a SQL query to find all the unique department in the employees table.

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

ORDER By Statement

Write a SQL query to retrieve all employees first and last names, ordered by Salary in descending order

```
SELECT first_name,
       last_name
FROM employees
ORDER BY Salary DESC;
```

First_name	Last_name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Sarah	Brown
Daniel	Clark
David	White

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LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employees.

```
SELECT *
FROM employees
ORDER BY Salary DESC
LIMIT 5;
```

Id	first_name	Lastname	department	Salary	hire_date	City
6	Emily	Davis	IT	62000	2015/02/14	Chicago
3	Mike	Johnson	Finace	60000	2017/09/30	LA
7	Robert	Wilson	Finance	59000	2019/10/01	Houston
1	John	Doe	IT	55000	2018/06/15	NY
4	Sarah	Brown	IT	53000	2021/03/25	NY

WHERE Statement

Write a SQL query to find employees who work in the IT department.

```
SELECT *
FROM employees
WHERE department = 'IT';
```

Id	First_name	Last_name	dep	Salary	hire_date	City
1	John	Doe	IT	55000	2018/06/15	NY
4	Sarah	Brown	IT	53000	2021/03/25	NY
6	Emily	Davis	IT	62000	2015/02/14	Chicago
10	Laura	Hall	IT	50000	2020/05/10	San Francisco

AND Statement

Write a SQL query to find employees who work in the finance department AND have a Sanary greater than 58,000

```
SELECT *
FROM employees
WHERE department = Finance AND
Salary > 58000;
```

Id	First_name	last_name	dep	Salary	hire_date	City
3	Mike	Johnson	Finance	60000	2017/09/30	LA
7	Robert	Wilson	Finance	59000	2019/10/01	Houston

OR Statement

Write a SQL query to find employees who work in the HR department OR the marketing departmet

```
SELECT *
FROM employee
WHERE department = 'HR' OR
department = 'marketing'
```

Id	First_name	Last_name	dept	Salary	hire_date	City
2	Jane	Smith	HR	48000	2019/07/20	Chicago
5	David	White	Marketing	52000	2016/04/10	San Francisco
8	Jessica	Moore	HR	51000	2018/05/22	LA
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago

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NOT Statement

Write a SQL query to find employees who do not work in the IT department

```
SELECT *
FROM employees
WHERE NOT department = 'IT';
```

Id	First_name	Last_name	dep	Salary	hire_date	City
2	Jane	Smith	HR	48000	2019/07/20	Chicago
3	Mike	Johnson	Finance	60000	2017/04/30	LA
5	David	White	Marketing	52000	2016/04/10	San Francisco
7	Robert	Wilson	Finance	59000	2019/10/01	Houston
8	Jessica	Moore	HR	51000	2018/05/22	LA
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago

IN Statement

Write a SQL query to find employees who are in the HR, IT, or Finance departments

```
SELECT *
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');
```

Id	First_name	Last_name	depart	Salary	hire_date	City
1	John	Doe	IT	55000	2018/06/15	NY
2	Jane	Smith	HR	48000	2019/07/20	Chicago
3	Mike	Johnson	Finance	60000	2017/04/30	LA
4	Sarah	Brown	IT	53000	2021/03/25	NY
6	Emily	Davis	IT	62000	2015/02/14	Chicago
7	Robert	Wilson	Finance	59000	2019/10/01	Houston
8	Jessica	Moore	HR	51000	2018/05/22	LA
10	Laura	Hall	IT	50000	2020/08/10	San Francisco

Combining Conditions

Write a SQL query to find employees who are in the IT department, have a Salary greater than 50000, and are located in New York

```
SELECT *
FROM employees
WHERE department = 'IT'
AND Salary > 50000
AND City = 'New York';
```

Id	First_name	Last_name	depart	Salary	Hire_date	City
1	John	Doe	IT	55000	2018/06/15	NY
4	Sarah	Brown	IT	53000	2021/03/25	NY

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Combining WHERE, AND, and ORDER BY

Write a SQL query to retrieve the first and last names of employees who work in the Finance or Marketing department, earn more than 52000 and order the results by Salary in descending order

```
SELECT first_name,
       last_name
FROM employees
WHERE department IN
      (Finance, marketing)
      AND Salary>52000
ORDER BY Salary DESC;
```

First_name	Last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

Combining SELECT DISTINCT, WHERE, and IN

Write a SQL query to find all the unique cities where employees work excluding in the IT and HR department

```
SELECT DISTINCT city
FROM Employees
WHERE department NOT IN
      ('IT', 'HR');
```

City
Los Angeles
San Francisco
Houston
Chicago

Combinig WHERE, NOT, AND, and ORDER BY

Write a SQL query to retrieve employees who are NOT in the Finance department, have a Salary greater than 50000 and order the results by hire date in Ascending order

```
SELECT *
FROM employees
WHERE department !=
      'Finance' AND Salary>
      50000
ORDER BY hire_date ASC;
```

Id	First_name	last_name	depart	Salary	hire_date	City
6	Emily	Davis	IT	62000	2015/02/14	Chicago
5	David	White	Marketing	52000	2016/04/10	San Francisco
8	Jessica	Moore	HR	51000	2018/05/22	LA
1	John	Doe	IT	55000	2018/06/15	NY
4	Sarah	Brown	IT	53000	2021/03/25	NY
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago

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Combining WHERE, OR, IN and LIMIT

Write a SQL query to find the first 3 employees who work in either Chicago or LA and belong to the IT or Marketing department

```
SELECT *
FROM employees
WHERE city IN ('Chicago', 'LA')
    AND department IN
    ('IT', 'Marketing')
```

Id	First_name	last_name	depart	Salary	hire_date	City.
6	Emily	Davis	IT	62000	2015/02/14	Chicago
9	Daniel	Clark	Marketing	53000	2022/06/01	Chicago